W-R-175/6A
22 December 1986
SUPERSEDING
W-R-175/6-T(GSA-FSS)
10 August 1970

FEDERAL SPECIFICATION SHEET

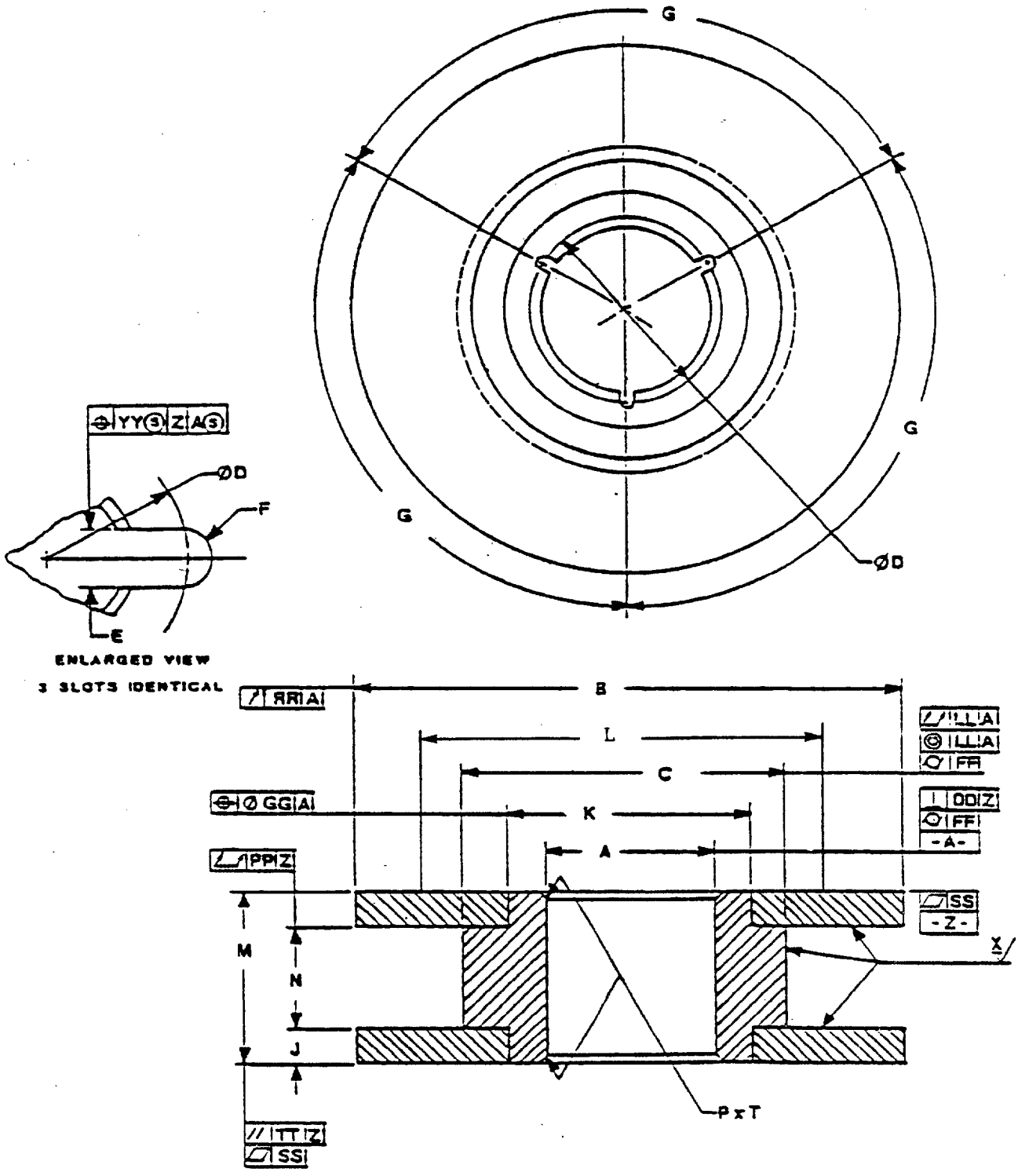
REELS, PRECISION, GLASS FLANGE WITH ALUMINUM HUB, 3-INCH CENTER HOLE

This specification sheet is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The requirements for acquiring the reels described herein shall consist of this specification and the latest issue of W-R-175.

AMSC N/A
FSC 5835
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SH 13202775

FIGURE 1. Reel configuration.

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REQUIREMENTS:

1. Materials: The reels shall be constructed with strengthened glass flanges bonded to a hub that is made of aluminum or an alloy with aluminum as the principal element.

CAUTION: The glass used in the construction of these reels is strengthened to withstand considerably more abuse than standard window glass; however, this glass can be broken if improperly used. The thickness of the coating shall be not less than 0.0012 (0.030 mm). The coating shall be transparent or translucent so that the quantity of tape on the reel is clearly visible to the unaided normal eye. A plastic coating has been specified to restrain glass particles in the event of fracture and to protect against scratches which tend to weaken the glass. The use of glass reels to provide improved protection for the tape involves the risk that improper handling may break the glass. For this reason, each reel shall carry the following caution:

"CAUTION - PRECISION GLASS REEL - HANDLE WITH CARE"

Lettering used for this statement shall be a minimum of 1/4 inch high and shall be affixed to withstand the temperature and humidity cycling specified in W-R-175.

2. Plastic coating: For purposes of safety, the glass flanges shall be provided with a plastic coating that will restrain glass particles in the event of flange fracture. The thickness of the coating shall be not less than 0.0012 (0.030 mm). The coating shall be transparent or translucent so that the quantity of tape on the reel is clearly visible to the unaided normal eye. Any such coating shall undergo no visible deterioration or delamination when the reel is temperature humidity cycled.

3. Flange fastening method: Flanges shall be secured to the hub by bonding with an adhesive. The bonded area shall be 2.25 square inches minimum for each flange. There shall be no visible excess of adhesive on the flanges or hub surfaces after completion of the curing cycle.

4. Flange fastening strength: The flanges shall remain securely attached to the hub after being tested in accordance with the test procedure defined in this specification sheet.

5. Flange wobble: The glass flange shall not contact the width gauge at any point on the inside surface of either flange when subjected to the flange wobble test in accordance with W-R-175.

6. Flange impact strength: The glass flange shall not break or crack under the impact momentum of 16 foot pounds per second when subjected to the ball drop test as defined in this specification sheet.

7. Tape threading: The requirement for a flange hole or cutout to facilitate tape threading as specified in W-R-175 shall not apply.

8. Threading method: The outside hub surface shall be provided with a coating that shall exhibit sufficient electrostatic charge to attract the tape to the hub for threading.

Reels and hubs shall be in accordance with the dimensional requirements of tables I, II and III (see figures 1 and 2).

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TABLE I. Dimensions.[1] [2]

Characteristic[3]	Reel and hub indicator	Dimensions		Flange deflection limit
		maximum	minimum	
Inside hub dia (A)		3.004 (76.302)	3.000 (76.200)	
Outside flange dia (B)	PGP	10.510 (266.95)	10.490 (266.45)	0.014 (0.356)
	QGP	12.510 (317.75)	12.490 (317.25)	.019 (0.483)
	SGP	14.010 (355.85)	13.990 (355.35)	.021 (0.533)
	TGP	15.010 (381.25)	14.990 (380.75)	.025 (0.635)
	UGP	16.010 (406.65)	15.990 (406.15)	.030 (0.762)
Outside hub dia (C)		4.510 (114.56)[4]	4.490 (114.05)	
Keyslot pitch dia (D)		3.252 (82.601)	3.248 (82.499)	
Keyslot width (E)		0.225 (5.715)	0.219 (5.563)	
Keyslot radius (F)		.5E	.5E	
Angle between successive keyslots (degrees) (G)		120.1 (2.096rad)	119.9 (2.093rad)	
Flange thickness (J)		0.095 (2.413)	0.085 (2.159)	
Inside diameter of flange diameter[5] (K)		3.700 (93.980)[6]	3.600 (91.440)[6]	
Flange clearance between lateral mounting surfaces, dia[11] (L)			6.0 (152.4)	

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Tape width indicator	Dimensions		Flange deflection limit
		maximum	minimum	
Overall reel width within reel width mounting surface (M)	250	0.465 (11.811)	0.459 (11.659)	
	500	.715 (18.161)	.709 (18.009)	
	750	.965 (24.511)	.959 (24.359)	
	1000	1.215 (30.861)	1.209 (30.709)	
	1500	1.715 (43.561)	1.709 (43.409)	
	2000	2.215 (56.261)	2.209 (56.109)	
Distance between flanges at hub (N)	250	0.272 (6.909)	0.268 (6.807)	
	500	.522 (13.259)	.518 (13.157)	
	750	.772 (19.609)	.768 (19.507)	
	1000	1.022 (25.959)	1.018 (25.857)	
	1500	1.522 (38.659)	1.518 (38.557)	
	2000	2.022 (51.359)	2.018 (51.257)	
Hub chamfer angle (degrees) (P)		46.0 (0.80rad)	44.0 (0.77rad)	
Hub chamfer (T)		0.040 (1.02)	0.020 (0.51)	
Mounting surface parallelism (TT)		.0014 (0.0356)		
Mounting surface flatness (SS)		.002 (0.051)		
Inside hub surface perpendicularity (DD)		.002 (0.051)[7]		
Inside hub surface cylindricity (FF)		.002 (0.051)[7]		

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Reel and hub indicator	Dimensions		Flange deflection limit
		maximum	minimum	
Outside hub surface cylindricity (FF)		0.002 (0.051)[8]		
Outside hub surface concentricity (total runout) (LL)		.002 (0.051)		
Flange rim concentricity (RR)		.020 (0.51)		
Flange wobble (PP)	PGP	.020 (0.508)		
	QGP	.020 (.508)		
	SGP	.020 (.508)		
	TGP	.025 (.635)		
	UGP	.030 (.762)		
Keyslot position (YY)		.005 (.127)		
Tape path clearance[9]				
Moment of inertia[10]				
Roughness of inside flange surface (X)		50 microinches		
Roughness of outside hub surface (X)		50 microinches		
Open flange area (percent)		0		
Diameter position (GG)		0.002 (0.051)		

See footnotes at top of next page.

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1 Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
2 All sharp edges shall be rounded to at least 0.004 (0.102) radius.
3 Where applicable, tolerance zones are indicated in parentheses.
4 This dimension is exclusive of any coating used on the reel hub. The coating shall meet the following requirements:

(a) The thickness of coating shall be 0.0012 (0.030) maximum.
(b) The outside diameter of the coated hub shall meet the cylindricity and concentricity requirements of the uncoated hubs.
(c) The coating shall meet the temperature and humidity requirements of W-R-175.

5 Lateral mounting surfaces extend out to "K" diameter.
6 The "K" diameter for the flange shall be 3.700 maximum, and the "K" diameter for the hub shall be 3.600 minimum. The "K" diameter is included for reference purposes only. The "K" diameter will not be tested.
7 The inside hub surface cylindricity and perpendicularity for reels for 0.250- and 0.500-inch tape shall be not greater than 0.0005 (0.0127).
8 The outside hub surface cylindricity for reels for 0.250- and 0.500-inch tape shall be not greater than 0.0008 (0.0203).
9 The tape path clearance shall be measured in accordance with W-R-175. The applicable tape path centerline for a given tape shall be as follows:

Tape width indicator	Tape path center line
250	0.229 (5.817)
500	.355 (9.017)
750	.480 (12.192)
1000	.605 (15.367)
1500	.855 (21.717)
2000	1.105 (28.067)

10 See table II.
11 "L" is the minimum diameter within which the flange, its fasteners and/or any labeling shall not protrude outside the planes of the lateral mounting surfaces.

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TABLE II. Moments of inertia.[1]

Tape width indicator	Moment of inertia (max)				
	PGP	QGP	SGP	TGP	UGP
250	0.16(6.7)	0.33(13.9)	0.49(20.6)	0.62(26.1)	0.79(33.3)
500	.16(6.7)	.33(13.9)	.49(20.6)	.63(26.5)	.80(33.7)
750	.17(7.2)	.34(14.3)	.50(21.1)	.65(27.4)	.82(34.6)
1000	.17(7.2)	.34(14.3)	.51(21.5)	.66(27.8)	.83(35.0)
1500	.18(7.7)	.35(15.0)	.59(24.8)	.69(29.6)	.85(36.4)
2000	.22(9.3)	.39(16.4)	.62(26.1)	.71(29.9)	.87(36.7)

[1] Units are pound-feet squared. Metric equivalents (gram-meter squared) are included for information only.

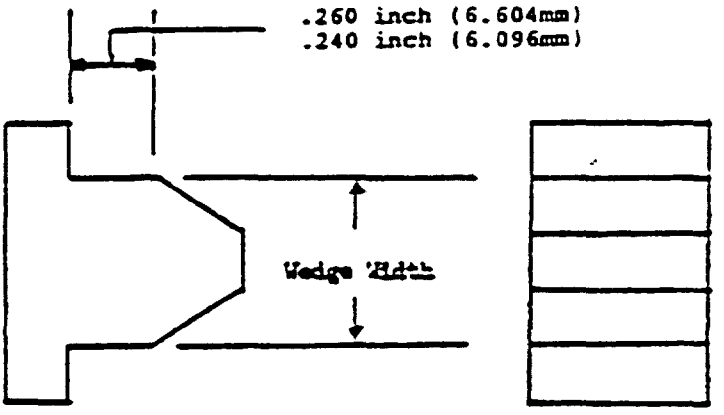
QUALIFICATION:

1. The qualification sample shall consist of three completely assembled type ER-1000-SGP reels; and three separate hubs (for the type ER-1000-SGP reels) prior to coating and assembly. Qualification of the sample reels ER-1000-SGP shall constitute qualification for all type VI reels regardless of width and flange diameter.

2. Quality conformance: The reel flange fastening strength test shall come under group A examinations and tests of W-R-175. The ball drop test shall come under group C examinations and tests of W-R-175.

3. Reel flange fastening strength test: The apparatus for this test shall consist of special wedge fixture made of a material which will not damage nor abraid the glass flange and shall be in accordance with figure 2.

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FIGURE 2. Wedge for flange fastening strength test.

TABLE III. Wedge width dimensions.[1]

Tape width indicator	Reel indicator				
	PGP	QGP	SGP	TGP	UGP
250	max 0.51(12.95)	0.59(14.99)	0.65(16.51)	0.69(17.53)	0.73(18.54)
	min .49(12.45)	.57(14.48)	.63(16.00)	.67(17.02)	.71(18.03)
500	max .76(19.30)	.84(21.34)	.90(22.86)	.94(23.88)	.98(24.89)
	min .74(18.80)	.82(20.83)	.88(22.35)	.92(23.37)	.96(24.38)
750	max 1.01(25.65)	1.09(27.69)	1.15(29.21)	1.19(30.23)	1.23(31.24)
	min 0.99(25.15)	1.07(27.18)	1.13(28.70)	1.17(29.72)	1.21(30.73)
1000	max 1.26(32.00)	1.34(34.04)	1.40(35.56)	1.44(36.58)	1.48(37.59)
	min 1.24(31.50)	1.32(33.53)	1.38(35.05)	1.42(36.07)	1.46(37.08)
1500	max 1.76(44.70)	1.84(46.74)	1.90(48.26)	1.94(49.28)	1.98(50.29)
	min 1.74(44.20)	1.82(46.23)	1.88(47.75)	1.92(48.77)	1.96(49.78)
2000	max 2.26(57.40)	2.34(59.44)	2.40(60.96)	2.44(61.98)	2.48(62.99)
	min 2.24(56.90)	2.32(58.93)	2.38(60.45)	2.42(61.47)	2.46(62.48)

[1] Dimensions are in inches. Metric equivalents are included for information only

Procedure: The wedge shall be inserted between the two glass flanges and allowed to remain there for a period of not less than 2 minutes. The test shall be conducted at four locations, approximately 90 degrees apart, on each reel. The reel shall then be inspected to verify that the flanges remain securely attached to the hub.

Flange wobble test: This test shall be the same as specified in W-R-175, except as follows:

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While the reel is still mounted in the concentricity fixture set up (see W-R-175), the applicable width gauge shall be mounted so that it is centered on dimension "N" and can be moved between the reel flanges in a straight line which is 90 degrees to the centerline between both lathe centers. The reel shall be rotated and the width gauge shall be moved between the flanges from the outside flange rim to the hub surface.

Ball drop test: Test reels shall be kept at ambient temperature and relative humidity conditions as specified in W-R-175, for a period of not less than 4 hours immediately preceding the test. The test reel shall be clamped securely in a horizontal position and a solid steel sphere that has a weight between 1.5 to 2 pounds shall be dropped freely and from rest, striking the reel flange within 1 inch of its periphery. The drop height shall be determined by the following formula:

$$S = \frac{M^2}{2m^2 g}$$

Where:

M = 16 foot pounds per second
m = ball mass in pounds
g = 32 feet per second per second
S = drop height in feet

The test shall be performed at two locations on the reel.

Hub measurements: The outside diameter of the hub, the outside hub surface cylindricity and the outside hub surface concentricity shall be measured on the coated and the uncoated reel hubs to determine conformance to the requirements specified herein (see table I).

MILITARY INTEREST:

Custodians

Army - CR
Navy - SH
Air Force - 99

Review activities

Navy - NV, OM
Air Force - 80
DLA - ES

User activity

Navy - AS

CIVIL COORDINATING ACTIVITY:
NASA - NPPO

Preparing activity:
Navy - SH
(Project 5835-0079-5)

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